

**HYDROPHILIX****DOCUMENT TITLE:**
Category II Supplier Assessment Form**Revision:**
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FM3-QM.HX015**Effective Date:**
25Sep2020

Category II Supplier Assessment Form

Complete the following information for the consultant or contractor in question:

Consultant or Contractor Name:	Sigi Caron
Title:	V. P. Regulatory / Clinical Affairs
Company (if applicable):	Biologics & Medical Device Consulting Group
Area of Service to be Provided:	☒ Quality Assurance ☒ Regulatory ☐ Manufacturing Operations ☐ Design and Development (R&D / Engineering) ☐ Software ☐ Clinical / Medical ☐ Marketing ☒ Other: Clinical Affairs
Specific Services to be provided:	Regulatory submissions; clinical evaluation; RA/CA/QA support
Curriculum vitae (CV) or Resume Attached?	☒ Yes ☐ No
The CV / resume has been reviewed and the contractor/consultant approved for the specific services to be provided:	☒ Approved ☐ Conditional ☐ Rejected / Disqualified
Comments/Additional Information:	
Approved By: (Name/Title/Date)	Lois Thomas / CEO / 04 November 2020

Sigi Caron
April 19 2022

∞ EXECUTIVE SUMMARY ∞

Over twenty-five years of progressive research, regulatory, quality, clinical and marketing experience in academic settings and in the medical device and pharmaceutical industry:

- **REGULATORY & CLINICAL AFFAIRS:** Gained FDA approval for multiple PMA's (modular and traditional); IDE's, IND's (drugs, cells [autologous, allogenic, embryonic stem cells], and gene products, standard, emergency and single patient), 510(k)'s; MAFs; *de novo* petitions; EUAs, HUD/HDEs; orphan designations; fast track designation, RMAT, Breakthrough Designations, developed preclinical and clinical studies; successfully submitted international registrations (incl. CE, Japan, Canada, and Australia); regulatory oversight of DSMB studies; implemented clinical auditing programs.
- **CRO SERVICES:** Managed stem cell, gene therapy, drug, class II and class III medical device studies, and IVD studies (5 gene therapy studies [CLI, DPN, ALS], 5 cardiac stem cell studies, 4 stroke studies, 2 drug studies, 3 cardiac ablation studies, 3 critical care diagnostic device study, 3 other diagnostic devices (2 neuro, 1 cancer), 2 orphan drug studies in a pediatric patient population for inborn errors of metabolism). Developed remote monitoring methods during COVID-19 pandemic. Currently overseeing 2 COVID-19 emergency studies
- **MEDICAL WRITING:** Clinical Evaluation Reports for EU submissions across multiple disciplines; ghost wrote 12 papers published in peer reviewed journals for gene and stem cell studies (diabetes, ALS, critical limb ischemia, cardiology, plasma adsorption)
- **QUALITY ASSURANCE:** Process validations; SPC strategies; successful management of FDA and ISO audits; product recalls; developed risk management programs (FMEA, CAPA, MDR integrated program), implemented multiple quality systems at medical device and biologics companies
- **MARKETING:** Managed a neuroendoscopic business, drug delivery company; wrote several business plans to obtain venture capital funding and SBIR; developed reimbursement strategies
- **SCIENCE:** 3 years as an analytical chemist; 1 year as a consultant; 3 peer reviewed publications; grant writing.

∞ EMPLOYMENT HISTORY ∞



2-02 - PRESENT

BIOLOGICS & MEDICAL DEVICE CONSULTING GROUP PRESIDENT / CEO

Head a small regulatory and clinical services consulting group (18 employees) with specialization in ICU, cardiovascular, neurointervention, and diabetes research. Serve multiple medical device, *in vitro* diagnostic, pharmaceutical and early stage biologics companies by providing regulatory consulting, including clinical protocol development, IDE, IND, emergency INDs, PMA, HDE, EUA and 510(k) submissions, device master files (MAF) *de novo* petitions,

orphan drug designations and HUDs, fast track designations, RMATs, breakthrough designations, interaction with FDA, interactions with NIH (for diagnostic and gene therapy studies), and development of worldwide regulatory strategies. Serve as CRO for phase I, II, and III studies and for COVID-19 emergency studies. Client companies in blood filtration / plasma adsorption, cardiology, electrophysiology, cancer treatments, wound healing, ophthalmology, orthopedic, neurology, neurointervention, ALS, diabetes monitoring, pediatric inborn errors of metabolism, general surgery and radiation devices, coatings and drug delivery platforms. Specific accomplishments include:

- Have been / currently am principal regulatory (FDA) representative for multiple companies:
 - Total artificial heart replacement company
 - International cardiac ablation company
 - Biologics Delivery Systems (a company in the Johnson & Johnson family of companies)
 - Embryonic stem cell company treating age related macular degeneration
 - One company specializing in RNA based diagnostics
 - One cancer diagnostic company
 - Once cancer therapeutics company
 - One cord blood-based therapeutics company
 - Two companies developing drugs to treat pediatric patients for inborn errors of metabolism
 - Two software development companies
 - Two cardiac mapping companies
 - Four neurointerventional companies
 - Four device companies in the diabetes space
 - One drug delivery device
 - Two intensive care device companies
- Obtained breakthrough designation for a breast cancer detection technology
- Obtained EUA for the first non-diagnostic / non-ventilator therapeutic medical device (EUA 200148) during the COVID-19 pandemic
- Developed new remote monitoring strategies specifically for 2 COVID-19 studies
- Successfully prepared and coached clients for pivotal face-to-face meetings with FDA (pre-IDE, pre-IND, INTERACT, pre-510(k), other Q-subs, end of phase II meetings, pre-PMA, HDE, and 100-day meetings)
- Successful negotiations with FDA to remove studies from clinical hold (gene therapy studies, and cardiac stem cell therapies)
- Successful submission of multiple INDs for phase I, II and III drugs and biologics (stem cells, gene products, drugs). INDs filed include standard, full protocol INDs, emergency INDs and single patient INDs. Disease states being studied under these approved INDs include heart failure, ischemic heart disease, heart transplant, peripheral diabetic neuropathy, ALS, critical limb ischemia, age related macular degeneration (AMD), diabetic ulcers, genetic disorders
- Successful submission of Orphan Drug Designation for a gene therapy product
- Successful submission of HUD for a cerebral vasospasm device
- Successful submission of a Fast Track Designation Request for a gene therapy product
- Successful submission of multiple IDEs (active implantables (cardiovascular)),

electrophysiology catheters, cardiac ablation systems, diagnostics, ophthalmic implants, continuous blood glucose monitors, stroke devices, liquid embolic, brachytherapy devices, intensive care unit devices)

- Successful submission of seven full PMAs (4 ablation catheters, 1 renal stent, 1 diabetic monitoring device, 1 device/biologic combination product through CBER)
- Successful submission of multiple PMA supplements (cardiovascular, burn management device, total artificial heart)
- Successful submission of 510(k)s and de novo submissions, including cardiovascular mapping and imaging systems, electrophysiology catheters, glaucoma shunts, lacrimal stents, endoscopes; diagnostic devices, refractometer, software applications, emergency medicine devices, general hospital devices, neurothrombectomy devices, peripheral thrombectomy devices, IVDs, and several dermatology devices
- Successful submission of class III Design Dossiers (cardiology, neurology, insulin pump, and electrophysiology)
- Developed multiple clinical protocols for cardiovascular, diabetic neuropathy, glucose monitoring, diagnostic devices, critical limb ischemia, ALS, aesthetic surgery, organ donation, neurology, liver failure, diagnostic studies for neurologic applications, device for treatment of sepsis, and ocular devices
- Management of stem cell studies (heart failure, phase I, phase II; age related macular degeneration, phase I/II)
- Management of gene therapy studies (CLI, phase II and phase III; diabetic peripheral neuropathy, phase I/II, phase II, and phase III; ALS phase I/II and phase IIb)
- Management of 4 neuro interventional studies
- Management of one drug study for pediatric patients with inborn errors of metabolism
- Ghost wrote 12 peer-reviewed publications for stem cell and gene therapy studies (diabetes, ALS, critical limb ischemia, cardiology, plasma adsorption, veterinary study)
- Business plan development for venture funding (trauma devices, drug coatings for cardiovascular devices, drug admixture, reconstructive surgery device, and anti-viral drug)
- Managed product recalls with minimal regulatory impact
- Managed FDA inspections and post-inspection follow-up
- Redesigned risk management programs

THERACARDIA INC.

2-00 – 2-02

VICE PRESIDENT RA/QA

6-01 – 2-02

DIRECTOR RA/QA

2-00 – 6-01

- Determined RA strategy for Class III cardiovascular devices
- Reduced clinical trial requirements (endpoint & enrollment requirements) for paradigm shift device by two thirds; helped shape FDA policy for CPR device
- Obtained CE mark approval 1 year early for novel class III cardiovascular device using a unique clinical review approach
- Put together a Data Safety Monitoring Board (DSMB) to oversee high risk clinical trial;
- Built and implemented a clinical auditing program
- Determined Medicare Reimbursement Strategy – set up an early IDE / national coverage meeting with CMS

SIGI CARON, BS, MBA, RAC

- Brought company into compliance with 21 CFR part 11 requirements
- Successfully navigated company through 2 ISO audits (including EN46001 certification); no observations noted
- Wrote business plan for C round funding, raising 12.5 million with a minimal dilution of existing shares
- Ghost wrote 2 papers for peer-reviewed journal

BIOSENSE WEBSTER, INC.

(DIVISION OF J&J)

REGULATORY AFFAIRS CONSULTANT

9-99– 2-00

- Determined RA strategies for electrophysiology & cardiology products. Specifically, joint venture drug/device and biologic/device submissions. Submissions included: modular PMA, IDE, MAF, special 510(k)
- Carried out recall on multiple product lines affected by packaging defect

SMITH & NEPHEW, INC. ENDOSCOPY DIVISION

1-98 – 9-99

RA MANAGER

9-98 – 9-99

RA/QA PROJECT MANAGER

1-98 – 9-98

- Determined RA and clinical strategies for products developed in-house, at the University of Massachusetts Smith & Nephew Development Center, and through a joint venture with Advanced Tissue Sciences (ATS, San Diego). Responsible for 2 facilities (160 personnel, 5 direct reports)
- Product portfolio included: biologics (tissue engineering), absorbable implants and novel implant delivery systems, other orthopaedic implants (knee, shoulder) and arthroscopic devices;
- Submissions included: 510(k)'s, IDE's, Modular PMA
- Re-engineered CE marking strategy for MDD / EN 46001 compliance; eliminated the need to generate separate technical files; reduced paperwork requirements by 95%
- Re-engineered design control systems to align with business process. Reduced paperwork / signatures by 50% and reduced cycle time on average by 30%. Design control System used as backbone for new business development process
- Co-led (with the Director of R&D and Director of Marketing) the New Business Development Team
- Re-engineered Customs / Specials process. Reduced cycle time by 50%
- Created and implemented a standard approach to risk analysis; set up training and an expert team of internal consultants to assist project teams;
- Developed a corrective and preventive action program for all 3 Endoscopy Division facilities
- Designed several animal studies to test the feasibility of ground breaking new technologies
- Developed a process validation procedure which incorporates SPC techniques
- ISO / QSR trainer for the company.

JOHNSON & JOHNSON PROFESSIONAL, INC.

1995 - 1998

ASSOCIATE PRODUCT DIRECTOR **1997-1998**

- Managed a \$1 million / year neuroendoscopic business
- Conducted market research and formed alliances with surgeon inventors;
- Obtained funding to restart new product development for an ailing product line; directed R&D efforts both in-house and through outside collaborations
- Set pricing / incentive programs to jumpstart sales. Trained sales force.

SENIOR RA SPECIALIST & INTERNET CONTENT **1996-1997**

LIAISON

RA SPECIALIST & INTERNET CONTENT **1995-1996**

LIAISON

ASSOCIATE RA SPECIALIST **1995**

- Obtained market clearance for US, European, and emerging Asian markets for neuro, spinal, trauma, and orthopaedic products (10)
- Supervised projects, training and evaluation of 3 RA associates
- Coordinated and implemented a CE marking strategy. Developed a worldwide database;
- Conducted 2 formal (FDA) product recalls
- Lead a team of 15 managers and engineers to merge design control SOP's (due to 2 large acquisitions); upon completion, gave company-wide training
- Streamlined change control procedures
- Conducted RA due diligence for potential technology licensing (tissue engineering)
- Designed and gave training for advertising, promotions, and sales leads on the Internet

CONSULTANT

1997-1998

- Wrote the quality policy manual for a start up company
- Did a QSR GAP analysis and rewrote affected SOP's for a small company

WOODS HOLE OCEANOGRAPHIC INSTITUTION

1991-1995

RESEARCH ASSISTANT III

1993-1995

RESEARCH ASSISTANT II

1991-1992

- Designed experiments, developed new chemical techniques, acquired computer and laboratory equipment, and organized 5 ocean going research cruises. Supervised one technician
- Designed and built a customized analytic system, saving our laboratory \$20,000
- Research resulted in three publications (one senior author) in highly prestigious scientific journals

✂ EDUCATION, PUBLICATIONS & AWARDS ✂

EDUCATION

MBA	(High Technology Program)	Northeastern University 1995
B.S. Biology	(emphasis in mathematics)	UCLA 1990

PEER-REVIEWED PUBLICATIONS

1. CARON, S. & RAINBOTH, W.J. Field analysis of color cues that elicit territorial behavior in the garibaldi, *Hypsypops rubicundus* (Pomacentridae). COPEIA, 1992 vol. 4.
2. JOHNSON, C.G., CARON, S., and BLOUGH, N.V. Combined HPLC/MS of the radical adducts of a fluorescamine-derivatized nitroxide. Analytical Chemistry, 1996 vol 68.
3. ANDREWS, S.S., CARON, S., BLOUGH, N.V., ZAFIRIOU, O.C. Photochemical oxygen consumption in marine waters: A major sink for colored dissolved organic matter? Limnology and Oceanography Vol. 45, No. 2, March 2000.

AWARDS

1986 - 1987	Full academic scholarship to Northland College
1988 - 1989	UCLA President's Undergraduate Fellowship
1989 - 1990	UCLA President's Undergraduate Fellowship

ADDITIONAL EDUCATION

- Harvard – 1 year graduate course in Immunology
- “From Proteins to Profits” – 10-week course sponsored by the Massachusetts Biotechnology Council
- Lead ISO Auditor
- Sales training (knee, hip, brain and spinal anatomy)
- GMP training
- Project Management Training